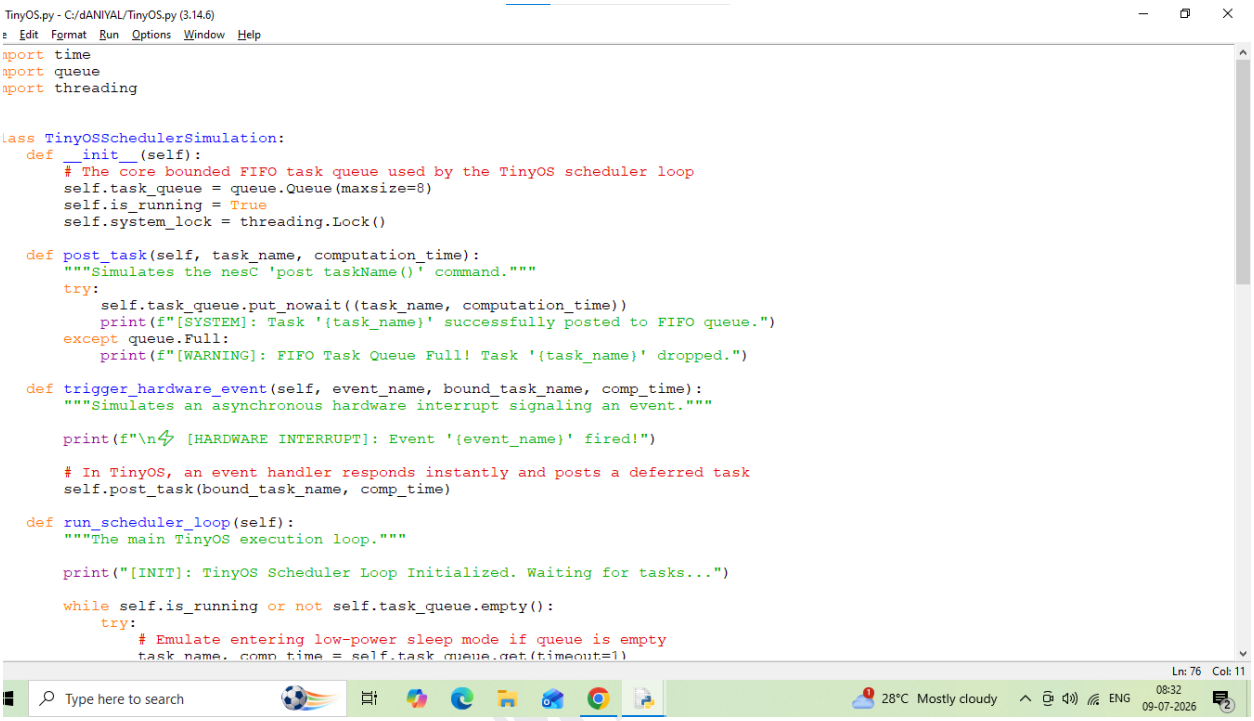
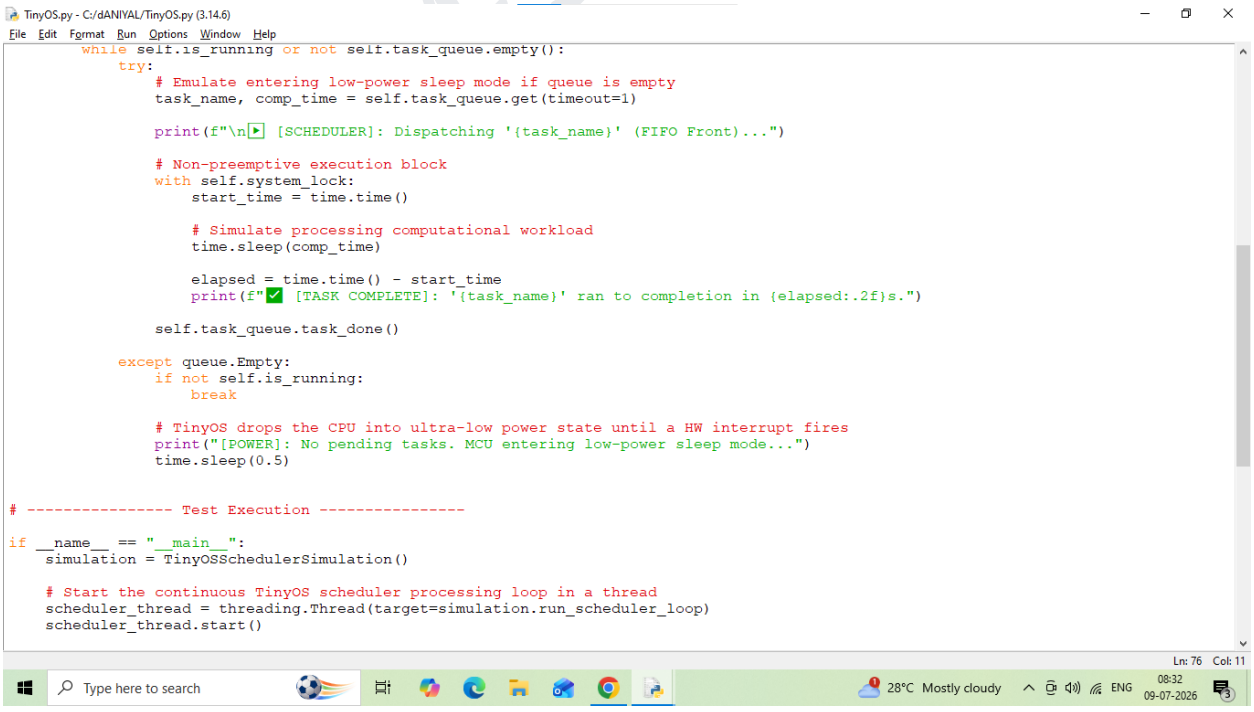
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Aim:-TinyOS utilizes a strict event-driven architecture coupled with a non-preemptive, First-In, First-Out (FIFO) task scheduler to minimize memory consumption and optimize power efficiency for resource-constrained wireless sensor networks

CODE:-





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TinyOS.py - C:/dANIYAL/TinyOS.py (3.14.6)

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```
if __name__ == "__main__":
    simulation = TinyOSSchedulerSimulation()

    # Start the continuous TinyOS scheduler processing loop in a thread
    scheduler_thread = threading.Thread(target=simulation.run_scheduler_loop)
    scheduler_thread.start()

    # Simulate random asynchronous hardware triggers
    time.sleep(0.5)
    simulation.trigger_hardware_event(
        "Timer0.fired",
        "ReadSensorTask",
        1.5
    )

    simulation.trigger_hardware_event(
        "Radio.receive",
        "ProcessPacketTask",
        0.8
    )

    # Fire another event while a task is running
    time.sleep(0.4)
    simulation.trigger_hardware_event(
        "ADC.dataReady",
        "FilterDataTask",
        0.5
    )

    # Allow simulation to finish
    time.sleep(4)

    simulation.is_running = False
    scheduler_thread.join()

    print("\n[STOP]: Simulation Finished.")
```

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OUTPUT:-

IDLE Shell 3.14.6

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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/dANIYAL/TinyOS.py ==========
[INIT]: TinyOS Scheduler Loop Initialized. Waiting for tasks...

 [HARDWARE INTERRUPT]: Event 'Timer0.fired' fired!

 [SCHEDULER]: Dispatching 'ReadSensorTask' (FIFO Front)...[SYSTEM]: Task 'ReadSensorTask' successfully posted to FIFO queue.


 [HARDWARE INTERRUPT]: Event 'Radio.receive' fired!
[SYSTEM]: Task 'ProcessPacketTask' successfully posted to FIFO queue.

 [HARDWARE INTERRUPT]: Event 'ADC.dataReady' fired!
[SYSTEM]: Task 'FilterDataTask' successfully posted to FIFO queue.
 [TASK COMPLETE]: 'ReadSensorTask' ran to completion in 1.50s.

 [SCHEDULER]: Dispatching 'ProcessPacketTask' (FIFO Front)...
 [TASK COMPLETE]: 'ProcessPacketTask' ran to completion in 0.80s.

 [SCHEDULER]: Dispatching 'FilterDataTask' (FIFO Front)...
 [TASK COMPLETE]: 'FilterDataTask' ran to completion in 0.50s.
[POWER]: No pending tasks. MCU entering low-power sleep mode...

[STOP]: Simulation Finished.
>>>
```

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